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$$x_n = \sqrt{n+1} - \sqrt{n}$$

$$\begin{aligned}\lim_{n \rightarrow \infty} \sqrt{n+1} - \sqrt{n} &= \lim_{n \rightarrow +\infty} \frac{n+1-n}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{1}{\infty} = 0\end{aligned}$$

$$\text{adh}_{n \rightarrow \infty} x_n = \{0\}$$

$$\limsup_{n \rightarrow +\infty} x_n = 0$$

$$\liminf_{n \rightarrow +\infty} x_n = 0$$

References